

Instructions:

- **Due April 7, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as **ENME403_S26_HW08_yourname.pdf**.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops conceptual understanding of root-locus methods for feed-back system analysis in the basic servo loop. Students will use geometric rules to construct root loci and interpret how open-loop poles and zeros shape closed-loop behavior, including systems with right-half-plane dynamics. They will determine gains that achieve specified damping ratios and identify stability limits through imaginary-axis crossings. Emphasis is placed on reasoning about system behavior from pole locations rather than relying on computation. MATLAB is used to visualize and confirm analytical predictions.

Problem 1. Consider the systems

$$1. G_1(s) = \frac{(s+2)^2}{(s+1)(s^2+1)},$$

$$2. G_2(s) = \frac{(s+1)(s+2)}{(s-1)(s-2)(s-3)}.$$

Assume that a proportional controller $G_c(s) = K_p$ is used in a basic servo loop.

- Using the root-locus rules, sketch the root locus by hand for each system.
- Using the MATLAB command `rlocus`, plot the root locus for each system.
- Indicate the point on the root locus at which the closed-loop poles have damping ratio $\zeta = 0.5$. Mark the intersection of the root locus with the $\zeta = 0.5$ line. Recall that a constant damping-ratio line satisfies

$$\theta = \cos^{-1}(\zeta) \quad (1)$$

measured from the negative real axis.

- Determine the gain K_p that corresponds to the closed-loop poles with $\zeta = 0.5$.

[**Hint:** Clicking on the MATLAB root locus plot displays the pole location, gain, frequency, and damping ratio.]

Problem 2. Consider the system

$$G(s) = \frac{s^4 + 10s^3 + 30s^2 + 10s - 51}{s^5 + 15s^4 + 86s^3 + 248s^2 + 368s + 192} \quad (2)$$

$$= \frac{(s-1)(s+3)(s^2+8s+17)}{(s+1)(s+4)(s+6)(s^2+4s+8)} \quad (3)$$

$$= \frac{(s-1)(s+3)(s+4-j)(s+4+j)}{(s+1)(s+4)(s+6)(s+2-2j)(s+2+2j)}. \quad (4)$$

1. Sketch the root locus of $G(s)$ by hand.
2. Use `pzmap` in MATLAB to plot the poles and zeros of $G(s)$.
3. Suppose a proportional controller $G_c(s) = K_p$ is used in a basic servo loop to modify the dynamic behavior of the system. Determine the largest value of the gain K_p such that the closed-loop system is asymptotically stable.
[Hint: The closed-loop poles lie on the imaginary axis when $1 + K_p G(s) = 0$ has purely imaginary solutions.]

Extra Credits Problem 1 (30 points.). In an experimental aircraft, the open-loop transfer function from the elevator deflection to the pitch angle is

$$G(s) = \frac{s - 1.5}{(s + 5.5)(s - 2)}. \quad (5)$$

1. Consider a proportional controller $G_c(s) = K_p$ in the basic servo loop (BSL).
 - (a) Sketch the root locus of the system by hand.
 - (b) Based on the root-locus geometry, determine whether there exists any value of K_p that stabilizes the closed-loop system.
 - (c) Briefly explain your answer using the location of open-loop poles and zeros.
2. Now consider the controller

$$G_c(s) = \frac{K_p}{s - p_c}, \quad (6)$$

where $p_c > 0$.

- (a) Explain qualitatively how adding the controller pole at $s = p_c$ modifies the root locus.
 - (b) For $p_c = 2.5, 3, 3.5$, sketch (or overlay using MATLAB) the root locus for each case.
 - (c) For each choice of p_c , determine whether a stabilizing gain K_p exists.
 - (d) Based on your observations, explain how the location of the controller pole affects stabilizability.
3. Select a value of p_c for which stabilization is possible.
 - (a) Determine a value of K_p that stabilizes the system.
 - (b) Report the corresponding closed-loop pole locations and comment on the expected transient behavior (e.g., oscillatory, slow/fast response). [Hint: In MATLAB, clicking on the root locus plot displays the pole location, gain, frequency, and damping ratio.]
 - (c) Using MATLAB, estimate the gain margin of the closed-loop system and briefly interpret its meaning in this context.